
Protein contacts are already in the attention: a single-forward-pass alternative to the Categorical Jacobian

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Abstract

The Categorical Jacobian of Zhang et al. [2024] reads protein contacts from a language model by perturbing every residue with every alternative amino acid, about $19L$ forward passes. We show the signal it reconstructs is already concentrated in a small subset of attention heads: averaging the top- K contact-relevant heads (selected on as few as 10 labeled proteins) recovers contacts in one forward pass and beats the Categorical Jacobian on leakage-clean data for every bidirectional model where logit-CJ is defined — and matches or beats it in-distribution across architectures, the lone exceptions being the smallest 8M model (where fusion trails at the shared $K=10$ default) and ESM Cambrian (a statistical tie). Ablations localize the gain to labeled head *selection*, not to averaging: at a matched label budget the unweighted mean ties a supervised L_1 logistic regression fit on the same heads, so the parameter-free mean is selection’s minimal form rather than the source of the advantage. Our primary test is leakage-clean. On a CAMEO split where neither selection nor evaluation touches data the models have plausibly memorized, the head readout beats the Categorical Jacobian on ESM-2-650M by +9pp ($N=29$, $p < 0.001$), with the within-model margin reproducing across architectures on a wider pretraining-aware set. Both methods fall 30–36 percentage points (fusion ~ 30 pp, CJ ~ 36 pp) from their in-distribution Zhang numbers to the leakage-clean numbers, a drop consistent with substantial pretraining overlap inflating prior in-distribution numbers; the CAMEO-vs-Zhang difficulty shift contributes as well, so we read it as an upper bound on the leakage component. We additionally introduce representation-CJ (repr-CJ), a hidden-state generalization of the Jacobian for architectures without a masked-LM head; show that the optimal K tracks how diffusely a model spreads its contact heads; and find that both methods lose the contact signal on both causal LMs we test (ProGen2-large and -xlarge, smaller no better than larger), suggesting attention-encoded pair structure may depend on bidirectional pretraining.¹

1 Introduction

Protein language models (PLMs) trained on hundreds of millions of unlabeled sequences learn rich representations of protein structure [Rives et al., 2021, Lin et al., 2023]. A natural question follows: can the structural knowledge a PLM acquires during pretraining be read out *without any supervised training on structure labels*? One answer, proposed by Zhang et al. [2024], is the Standard Categorical Jacobian (CJ): perturb each residue position with every alternative amino acid, measure the response in the model’s per-position masked-LM logits, and read residue–residue contacts off the resulting Jacobian. CJ reaches $P@L \approx 0.80$ on a 1,431-PDB-chain benchmark using ESM-2-650M

¹Code, figure-regeneration scripts, and the summary result tables (per-cell means and bootstrap intervals) are available at <https://github.com/Rome-1/plm-contact-fusion>.

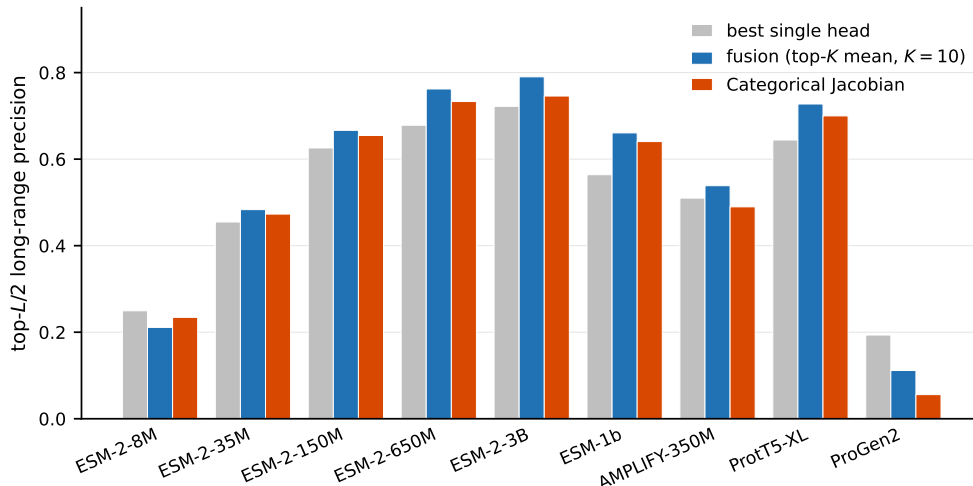


Figure 1: Top- $L/2$ long-range precision on Zhang eval-200 across nine protein-LM variants (ESM-2 at five scales, ESM-1b, AMPLIFY-350M, ProtT5-XL, ProGen2-xlarge). Naive-mean fusion of the top- K attention heads (blue) beats the per-protein top-1 head (grey) and the best-available Categorical Jacobian (red) on every cell at and above the 35M scale; the concentrated-cluster ESM-2-8M (best read at $K=3$, not the $K=10$ plotted here) is the in-distribution exception, and ProGen2-xlarge is the causal-LM scope boundary. “Best-available CJ” is logit-CJ [Zhang et al., 2024] where defined (ESM-2 family, ESM-1b) and repr-CJ (Section 3.2) where logit-CJ does not apply (AMPLIFY, ProtT5, ProGen2); the scope boundary is detailed in Section 4.4. Per-protein 95% bootstrap intervals in Appendix Table A6.

and requires no contact labels at all. However, the construction takes $\sim 19L$ forward passes for a protein of L residues (one perturbation per residue $\times$ 19 alternative amino acids) and is structurally tied to a per-position categorical head, which restricts it to bidirectional masked-LM architectures.

This paper offers a simpler answer: the same contact signal already sits in the attention maps of the model, concentrated in a small, identifiable subset of heads. Intuitively, residues that are close in three-dimensional space exert strong evolutionary constraints on each other — substituting one often requires a compensating change at the other to preserve the fold. Masked-LM pretraining captures these pairwise dependencies implicitly, and the attention mechanism is the natural site for the model to represent which positions “attend to” which. Bhattacharya et al. [2021] showed on ProtBERT-BFD that 6 of 480 attention heads carry most of the contact-relevant signal; Rao et al. [2021a] reached a related conclusion on ESM-1b via an L_1 -sparse supervised regression that effectively selects a small subset of heads, and found that simply averaging those heads unweighted outperforms the fitted regression. What is new here is not the mean — Rao et al. [2021a] already showed it beats their fitted regression — but that it needs no regression and no per-pair labels (a 10-protein ranking suffices), survives a leakage-clean evaluation, improves on the Categorical Jacobian, and generalizes to encoder-decoder and causal architectures. We extend this line in a label-efficient direction. The unweighted arithmetic mean of the top- K heads, with K chosen once from a disjoint 10-protein head-precision ranking and no per-pair weight fitting, gives an unsupervised contact-prediction signal that exceeds CJ on every masked-LM and encoder-decoder PLM we test, in one forward pass instead of $\sim 19L$, on data filtered to remove sequence-level pretraining overlap. The optimal K varies between architectures and tracks the diffuseness of the contact-head cluster: a mechanistic observation about how different PLMs allocate pair structure across their attention heads.

Contributions. The end-to-end leakage-clean comparison is the central finding (*leakage-clean*: evaluated only on protein chains deposited after a tested model’s pretraining cutoff and filtered for sequence similarity per Hermann et al. [2024]): with head selection on one half of CAMEO-PTA25 (a pretraining-aware CAMEO split of post-2022-09 targets, defined in Section 3.1) and evaluation on the disjoint other half (no Zhang chains anywhere in the pipeline), fusion on ESM-2-650M beats CJ, and CJ no longer beats the best single attention head (Section 4.1). The pattern reproduces

on a wider CAMEO-PTA25 set across three architectures (ESM-2-650M, ESM-2-3B, ProtT5-XL). Three further results follow. (i) *Cross-architecture replication* on Zhang eval-200 and CASP14-FM extends to five PLMs spanning masked-LM and encoder-decoder objectives. (ii) *Representation-CJ*, a hidden-state generalization of logit-CJ, applies to architectures lacking a per-position MLM head and agrees with logit-CJ where both are defined (Section 4.2). (iii) *Causal-LM scope*: both fusion and CJ collapse on both causal LMs we tested (ProGen2-xlarge and ProGen2-large), consistent with bidirectional pretraining being important for attention-encoded pair structure — though with five architectures and several correlated confounds we report this as associational, not causally identified (Section 4.4). Both methods drop 30–36pp absolute from Zhang eval-200 to leakage-clean CAMEO, upper-bounding how much of the prior unsupervised-contact-prediction headline reflects sequence overlap between pretraining and evaluation (a CAMEO difficulty shift contributes too).

2 Related Work

Unsupervised contact prediction from protein language models. Zhang et al. [2024] introduced the Standard Categorical Jacobian: perturb each residue with every alternative amino acid, measure the per-position categorical-distribution response, and read contacts off the resulting Jacobian. Their primary result is $P@L = 0.80$ on a 1,431-PDB-chain benchmark with ESM-2-650M, above an inverse-covariance baseline of 0.61. We adopt CJ as the direct unsupervised baseline this paper compares against and we reproduce it at $P@L/2 \text{ long} = 0.733$ on our 200-protein eval draw (head/ K selected on a disjoint 10-protein set) of their benchmark (the metric switch is for direct comparability with the attention-readout literature [Rao et al., 2021a,b], which reports top- $L/2$ long-range precision). Our fusion method extends this unsupervised-contact-prediction line — read contact signal off a pre-trained PLM without a supervised head — but reads attention rather than perturbing the input, exchanging $\sim 19L$ forwards for 1 forward at higher precision. Representation-CJ (next paragraph) is a parallel methodological piece.

Attention as a contact signal. The structural localization of pair information in transformer attention was first observed in NLP [Vig, 2019, Clark et al., 2019] and extended to PLMs by Vig et al. [2021], Bepler and Berger [2019], Rao et al. [2021a], Rao et al. [2021b], and Bhattacharya et al. [2021]. The closest methodological neighbor is Rao et al.’s L_1 -regularized logistic regression over per-head APC’d attention maps on ESM-1b, which acts as an implicit sparse head selector ($\sim 102/660$ heads receive non-zero weight; top- $L/2$ long-range = 0.533); they additionally report that averaging the top-10 selected heads *unweighted* outperforms the fitted regression. Our setup is closest in mechanism to that unweighted top- k variant: we differ in selecting on a small head-precision ranking (as few as 10 proteins) rather than via a regression fit to per-pair contact labels on full PDB chains. Rao et al. [2021b] extends the recipe to MSA inputs at higher precision. Bhattacharya et al. [2021] demonstrated, on *ProtBERT-BFD*, that contact signal concentrates in a small subset of heads (6 of 480) and that a single layer’s attention is often sufficient; the head-concentration phenomenon transfers to the ESM family in our cross-architecture sweep (Section 4.3). This prior work raises two questions we take up in the results: whether the gain comes from head *selection* or from averaging *per se*, and whether a fitted regression beats an unweighted mean at a matched label budget. Both are answered in Section 4.1 and Appendix G (in short: selection is the operation that matters, and at a 50-protein budget the fitted regression and our naive mean are statistically indistinguishable).

Mechanistic interpretability of PLMs. A recent line of work uses sparse autoencoders and circuit analysis to interpret what protein language models compute [Adams et al., 2025, Simon and Zou, 2025], identifying features for binding sites, structural motifs, and functional domains in ESM-2 embeddings. These papers preserve CJ as their accuracy metric and ask which model latents drive it. Our work is methodological: we move accuracy and compute. The “head cluster” structure we exploit in Section 4.3 is consistent with mech-interp findings that contact information localizes to a small set of attention heads.

Leakage as a confound in PLM evaluation. Hermann et al. [2024] showed that splits drawn before the pretraining cutoff inflate measured PLM downstream-task performance by 11.1% absolute on thermostability prediction; a pretraining-aware split (filtering test sequences against pretraining cluster representatives) recovers the model’s true generalization. Bartoszewicz et al. [2025] extend the same finding to protein-protein interaction inference. Earlier PLM benchmark suites TAPE

[Rao et al., 2019] and FLIP [Dallago et al., 2021] adopted identity-based holdout splits in the same spirit, and Detlefsen et al. [2022] flagged pretraining-data overlap as a confound for hidden-state-based PLM readouts more broadly. Anonymous [2025] introduces a contamination-free benchmark whose test set is restricted to proteins released after 2024. The structural-prediction literature has its own temporal-holdout protocols: CASP rounds release targets after the relevant pretraining cutoffs [Kryshtafovych et al., 2021], and AlphaFold2’s training/test split is chronologically separated by design [Jumper et al., 2021]. What is novel in our application is the specific Hermann-filter evaluation of the CJ-vs-attention-readout comparison for unsupervised contact prediction, with parameters chosen to match Hermann et al.’s sequence-similarity protocol (Section 4.1).

Generalization across architectures. Zhang et al. [2024] evaluate CJ exclusively on the ESM family. We extend the evaluation to an encoder-decoder model (ProtT5-XL; Elnaggar et al., 2021), bidirectional MLM variants beyond ESM-2 (ESM-1b, Rives et al., 2021; AMPLIFY-350M, Fournier et al., 2024), and a causal LM (ProGen2-xlarge; Madani et al., 2023); we introduce repr-CJ as the technical change that makes the cross-architecture comparison possible. We additionally evaluate ESM Cambrian (ESMC; ESM Team, 2024), the bidirectional masked-LM successor to ESM-2, distributed open-source as part of a public protein “world model” release [Biohub, 2026]: at both scales it exposes a categorical logit head and per-head attentions, so logit-CJ and fusion apply unchanged, and the within-model fusion-over-CJ margin holds on it (Section 4.1, Table 4). We leave the multi-track generative ESM-3 [Hayes et al., 2025] — whose generative architecture departs from the masked-LM readout our methods assume — to future work.

Supervised alternatives. Supervised structural prediction [Jumper et al., 2021, Lin et al., 2023] reaches substantially higher absolute contact precision than any unsupervised attention-readout method, including ours. The comparison is orthogonal: supervised methods learn a structure-prediction function on labeled contact maps, while unsupervised methods read out what the PLM has *already* encoded from sequence pretraining alone. The question we ask here — what does the model know without further training? — has a different epistemic status than the question of how accurately a supervised head can be fit on top of it. We comment in Section 5 on the implicit comparison.

3 Methods

3.1 Notation and protocol

A protein of L residues folds into a three-dimensional structure in which residues far apart in the chain can be close in space. A **contact** is a pair of residues (i, j) within $8 \text{ \AA}$ in the experimentally determined structure.² We restrict to *long-range* contacts ($|i - j| \geq 24$ along the chain), which are the most informative for fold determination and the hardest to predict from sequence alone [Dunn et al., 2008]. The contact map is the binary symmetric matrix $C \in \{0, 1\}^{L \times L}$ with $C[i, j] = 1$ for each long-range contact. Every method is evaluated by **top- $L/2$ long-range precision** ($P@L/2$ long): predict the $L/2$ highest-scoring long-range pairs, report the fraction that are real contacts [Zhang et al., 2024]. All scores rank from a single (L, L) score map per method.

We use three evaluation sets, summarized in Table 1. CAMEO-PTA25 is our primary, leakage-clean set: 553 single-chain CAMEO targets deposited on or after 2022-09-01 (post-dating ESM-2’s UniRef50 2021_04 pretraining release), filtered against current UniRef50 [Suzek et al., 2007] at 50% MMseqs2 [Steinegger and Söding, 2017] identity / 80% query coverage following the leakage protocol of Hermann et al. [2024]; of the 56 targets that pass, 39 retain $L \geq 75$ (the long-range eligibility cutoff), of which 29 form the leakage-clean evaluation half. The PTA25 tag marks the pretraining-aware CAMEO snapshot (assembled in 2025) rather than asserting cleanliness outright, because leakage cleanliness is model-relative. CAMEO is a moving target, and this split is leakage-clean only for models whose pretraining predates the 2022-09 deposition floor (e.g. ESM-2, ESM-1b, ProtT5). Models trained later, namely ESMC and AMPLIFY (both 2024), may have seen targets in

²We follow the standard criterion of each source benchmark: an $8 \text{ \AA}$ cutoff on the C_β atoms for the Zhang benchmark, following Rao et al. [2021a] (the C_α atom substitutes for glycine, which has no C_β), and an $8 \text{ \AA}$ C_α cutoff for CAMEO-PTA25. Zhang et al. [2024] instead use a looser $10 \text{ \AA}$ C_α cutoff and report precision over the top L pairs rather than the top $L/2$, so our absolute precisions are not directly comparable to the numbers in their paper; the comparison we rely on throughout is the gap between methods under one fixed criterion.

the post-2022-09 window, so we treat them as *partial* (AMPLIFY in Table 3; ESMC’s within-model CAMEO margin in Section 4.1).

Table 1: **Evaluation sets.** Selection budget is the disjoint labeled subset on which the top- K heads are picked; it never overlaps the evaluation draw. The primary, leakage-clean comparison (Section 4.1) runs on CAMEO-PTA25; cross-architecture reproduction numbers (Appendix Table A6) and the methodology analyses (operator comparison, K -sweep, cluster-diffuseness, depth probe, timing) run on Zhang eval-200. CASP14-FM lists the 16 nominal free-modeling targets; 14–16 are scored per model (a few long targets are dropped at load on individual cells), with the exact per-cell N in Table A6.

Dataset	Source	N (eval)	Sel. budget	Role
Zhang eval-200	Zhang benchmark [Zhang et al., 2024]	200	select-10	in-distribution repro.
CAMEO-PTA25	CAMEO + Hermann filter	29	select-10	leakage-clean primary
CASP14-FM	CASP14 [Kryshtafovych et al., 2021]	16	—	hard OOD

Zhang eval-200 is a 200-protein draw stratified by sequence length from Zhang et al.’s 1,431-PDB-chain benchmark, with head and K selection on a *disjoint* 10-protein set (Zhang select-10) so selection and evaluation never share a chain — the in-distribution analogue of the leakage-clean CAMEO split. CASP14-FM comprises 16 free-modeling targets of CASP14 [Kryshtafovych et al., 2021] (mean $L=423$, range 95–2180), a harder, longer test. The only 50-protein Zhang draw (*Zhang-50*) is the matched-budget supervised control in Appendix G, the larger budget the truncated-Rao L_1 -LR probe needs to fit; the selection-size study (Appendix N) uses the same pool and shows 10 labeled proteins suffice.

3.2 Standard CJ and representation-CJ

Standard CJ [Zhang et al., 2024] — which we also call **logit-CJ**, because it reads the model’s masked-LM logit output — builds a four-index Jacobian $J[i, a', j, a] = \text{logits}^{i \rightarrow a'}[j, a] - \text{logits}_{\text{unpert}}[j, a]$, where the superscript $i \rightarrow a'$ denotes the forward pass on the sequence with position i substituted to amino acid a' , and $\text{logits}_{\text{unpert}}$ is the native (unperturbed) forward pass. The two amino-acid indices play distinct roles: a' is the *substituted* amino acid placed at i , and a is the *response* amino acid whose predicted logit is read at every other position j . So $J[i, a', j, a]$ records how perturbing i to a' moves the logit for a at j , relative to the native sequence. Each position is perturbed to its 19 non-native amino acids, and one shared forward pass supplies the native baseline — so the per-protein cost is $19L + 1$ forwards, not $20L$ (the native is the baseline, not a perturbation). Standard preprocessing then collapses J to an (L, L) contact map: mean-center over the index axes, take the Frobenius norm over the two amino-acid axes a', a , zero the diagonal, apply average-product correction (APC; Dunn et al., 2008, which subtracts the per-row and per-column background co-variation that otherwise swamps the true contact signal), and symmetrize.

Logit-CJ requires a per-position categorical head, so it applies only to bidirectional masked-LM architectures (ESM, AMPLIFY). We generalize it to any architecture exposing per-position hidden states h (the residual-stream vector at a position and layer) by replacing the logit response with a hidden-state response: $J_{\text{repr}}[i, a', j] = \|h^{i \rightarrow a'}[\ell, j] - h_{\text{unpert}}[\ell, j]\|_2$, the distance the hidden state at position j , layer ℓ (default = last; the layer sweep in Appendix J confirms the last layer is the right default) moves when i is perturbed, in place of how the logits move, averaged over a' and given the same post-processing as logit-CJ. We call this **repr-CJ**; it applies to encoder-only, encoder-decoder, and causal models, at the same $19L + 1$ -forward cost. On causal models we additionally replace the symmetrize step with an upper-triangle-only readout (**causal-CJ**). The two give an identical top- $L/2$ long-range ranking: causal masking populates only one triangle of the response (a position responds only to its left context), so symmetrizing merely halves those entries relative to reading the populated triangle directly — a monotone rescaling that leaves the rank order of pairs, and hence top- $L/2$ long-range precision, unchanged.

3.3 Top- K attention fusion

For each (layer, head) of an attention-LM we extract the per-head attention map (symmetrize, APC, zero-diagonal) and score it by P@ $L/2$ long against a small labeled selection set, disjoint from the

evaluation set (both the Zhang eval-200 and the CAMEO-PTA25 results select on a disjoint 10-protein subset; only the matched-budget control against the L_1 -LR in Appendix G uses 50, the larger budget needed to fit that regression). A selection-set-size sweep (Appendix N) shows 10 proteins suffice. Averaging precisions gives a global head ranking; the **top- K heads** are the only labeled step. Our default operator is the arithmetic mean over the top- K APC’d maps: $S = \frac{1}{K} \sum_{k=1}^K A_k$, scored by $P@L/2$ long. Per-test cost: **one forward of the length- L sequence**.

We sweep $K \in \{1, 2, 3, 5, 7, 10, 20\}$ (Section 4.3) and ablate the arithmetic mean against nine alternative operators (elementwise max, geometric mean, z-score mean, reciprocal-rank fusion, positional-filter mean, entropy-gated mean, graph closure, modern-Hopfield, spectral consensus; Appendix E). The arithmetic mean is the per-cell best on 8 of 16 cells and within 0.3pp of the best alternative on most others; we adopt it as the cross-architecture default for simplicity. An APC ablation (Appendix F) confirms the fusion gain over the top-1 head comes from averaging, not from APC.

3.4 Models

We evaluate nine protein-LM variants spanning masked-LM (ESM-2 at five scales — 8M / 35M / 150M / 650M / 3B; ESM-1b; AMPLIFY-350M), span-corruption encoder-decoder (ProtT5-XL), and autoregressive causal-LM (ProGen2-xlarge) objectives. Per-model parameter counts, position encodings, pretraining corpora, and checkpoint identifiers are in Appendix D–C. Whether CJ’s contact-prediction result is a property of the Jacobian readout or of ESM-2 specifically was not addressed in the original work [Zhang et al., 2024]. The Categorical Jacobian has since been applied to other masked-LM and genomic sequence models (e.g. ESM3, Evo, gLM2), but, to our knowledge, not to an encoder-decoder or a causal protein language model; the evaluation below is, to our knowledge, the first such cross-objective test. Significance is assessed by paired bootstrap on per-protein $P@L/2$ long deltas (Appendix B); code, configurations, and the summary result tables are released with the paper.

4 Results

4.1 Fusion of top- K attention heads beats CJ across architectures

Any single-set claim involving PLMs is open to the criticism that the model has memorized the evaluation set during pretraining [Hermann et al., 2024]. We therefore lead with a leakage-clean readout, treat the pretraining-aware CAMEO set as the primary evaluation, and use Zhang eval-200 [Zhang et al., 2024] to reproduce numbers on a comparable benchmark.

Primary result: end-to-end leakage-clean. The comparison the paper rests on removes pretraining leakage end to end. On the pretraining-aware CAMEO-PTA25 split (Section 3.1: 56 of 553 post-cutoff targets survive the Hermann et al. [2024] leakage filter, only $\sim 10\%$, a measure of how leakage-prone contemporary CAMEO is, and 39 retain $L \geq 75$; construction and the conservative current-UniRef50 substitution in Appendix A), neither head selection nor evaluation touches data the models have plausibly memorized.

The cleanest comparison runs head selection and evaluation on *disjoint halves* of CAMEO-PTA25, so no Zhang chains appear anywhere in the pipeline. We select the top- K heads on a length-stratified 10-protein subset of the $L \geq 75$ CAMEO-PTA25 targets and evaluate fusion on the disjoint remaining $L \geq 75$ targets ($N=29$). This uses a 10-protein selection budget (matching the in-distribution protocol) rather than the larger half-split, which leaves more targets for evaluation. On ESM-2-650M, fusion beats Standard CJ significantly (Table 2); even the per-protein top-1 head already matches CJ (0.396 vs 0.369, n.s.), so on leakage-clean data CJ’s $\sim 19L$ perturbations buy nothing over reading a single head; fusion (0.459) then beats CJ by +9pp and the top-1 head by +6pp. We fix $K=10$ (the cross-architecture default).

Pretraining-aware cross-architecture. Running the same end-to-end leakage-clean protocol — head selection on a disjoint CAMEO select-10 subset, evaluation on the remaining $N=29$ targets, so no Zhang chains touch the pipeline — reproduces the pattern across architectures (Table 3): fusion beats CJ by +9.0, +9.4, +5.9, and +5.9pp on ESM-2-650M, ESM-2-3B, ProtT5-XL, and

Table 2: **End-to-end leakage-clean comparison on CAMEO-PTA25** ($L \geq 75$, $N=29$). Top- K heads selected on a disjoint 10-protein subset; per-protein paired bootstrap, $B=10,000$. Asterisks: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Method (ESM-2-650M)	P@L/2 long	Δ vs CJ
Per-protein top-1 head	0.396	+0.027 (n.s.)
Standard CJ	0.369	—
Fusion ($K=10$, naive-mean)	0.459***	+0.090***

Table 3: **CAMEO-PTA25 cross-architecture comparison (end-to-end leakage-clean)**. P@L/2 long on CAMEO-PTA25 ($L \geq 75$); top- K heads selected on a disjoint length-stratified 10-protein subset (CAMEO select-10) and evaluated on the remaining $N=29$ targets, so no Zhang chains appear anywhere in the pipeline. $K=10$ throughout (the cross-architecture default). Asterisks on Δ are paired-bootstrap significance (*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$). AMPLIFY-350M shows fusion vs. top-1 head only (it has no masked-LM logit head for logit-CJ); ProGen2-xlarge is treated as the causal-LM scope boundary in Section 4.4. AMPLIFY-350M is marked *partial* for CAMEO-PTA25 leakage cleanliness (Appendix D).

Architecture	K	top-1 head	fusion	CJ	$\Delta(\text{fusion} - \text{CJ})$
ESM-2-650M	10	0.396	0.459	0.369 (logit)	+0.090***
ESM-2-3B	10	0.347	0.463	0.369 (logit)	+0.094***
ProtT5-XL	10	0.354	0.418	0.359 (repr)	+0.059***
ESM-1b	10	0.340	0.348	0.289 (logit)	+0.059***
AMPLIFY-350M [†]	10	0.255	0.262	—	—

[†] partial leakage cleanliness: AMPLIFY pretraining post-dates the 2022-09 CAMEO deposition cutoff, so its absolute precision is not leakage-clean; the within-model $\Delta(\text{fusion} - \text{CJ})$ largely controls for it (both read the same model).

ESM-1b (all $p < 0.001$, paired bootstrap over $N=29$). These four comparisons read the *same* 29 evaluation proteins with different models, so they are correlated reads of one small draw rather than four independent replications; the consistent within-model sign of the margin is the robust signal, not four separate significance tests. These clean margins are *larger* than the partial-clean operating point we reported previously (selection on a contaminated corpus gave +7.2/+7.4/+5.2pp): clean selection removes contamination that had inflated CJ, widening the gap. For AMPLIFY-350M, which has no masked-LM logit head, we report fusion vs. top-1 only (+0.65pp); ESM-1b’s fusion-vs-top-1 margin is +0.8pp. On these clean data, ESM-2-3B edges ESM-2-650M only marginally (0.463 vs. 0.459), far below the in-distribution scaling gap (3B leads 650M by ~ 2.8 pp on Zhang eval-200): the larger model’s in-distribution edge is partly the memorization the Hermann filter removes.

Zhang eval-200 reproduction and length scaling. On the in-distribution Zhang eval-200 (direct reproduction comparability with Zhang et al. [2024]), fusion beats CJ on every architecture at and above the 35M scale with paired-bootstrap significance at $p < 0.01$ (Figure 1; per-cell numbers in Appendix Table A6). Two in-distribution exceptions. On the smallest ESM-2-8M the $K=10$ default over-selects a concentrated-cluster model (best read at $K=3$, Section 4.3) and fusion trails CJ (0.211 vs 0.234). On the later-added ESMC, fusion and logit-CJ are within noise (+0.004 on 300M; the gap opens to a significant +0.06–0.08 only on the harder leakage-clean CAMEO split, Table 3). On the harder, longer CASP14-FM (mean $L=423$) the gap widens, and *fusion on the smaller 650M exceeds CJ on the larger 3B* — a length-scaling pattern consistent with, though not proving, greater length sensitivity in CJ’s perturbation-based readout relative to fusion’s single forward. Per-architecture top- K values identified on the selection set transfer to a disjoint held-out draw without re-tuning (Appendix M).

For absolute calibration: ESMFold’s PAE contacts [Lin et al., 2023] reach P@L/2 long ≈ 0.6 – 0.8 on contemporary contact-prediction benchmarks; supervised MSA-augmented attention readouts [Rao et al., 2021b] reach ≈ 0.6 – 0.7 . Our leakage-clean operating point (≈ 0.4 absolute) sits below both, as expected for an unsupervised, MSA-free method on data the model has not memorized. The interpretable comparison is the +5-pp range gap over CJ.

A recent open backbone: ESM Cambrian. The central result holds on a recent open architecture: fusion beats the Categorical Jacobian on ESMC, and its within-model margin matches what ESM-2 showed two model generations earlier. ESMC [ESM Team, 2024, Biohub, 2026] is the bidirectional masked-LM successor to ESM-2, and we apply logit-CJ and fusion to it without modification. On in-distribution Zhang eval-200, ESMC’s higher contact scores at smaller parameter counts coincide with its stronger backbone: logit-CJ exceeds the same-tier ESM-2 baseline (both ESMC bootstrap CIs lie above the ESM-2 point estimate) and fusion roughly matches it, at $2\text{--}5\times$ fewer parameters (per-tier numbers in Table 4).

These absolute gains carry a bound. ESMC’s 2024 pretraining post-dates both Zhang eval-200 and our 2022-09-cutoff CAMEO-PTA25 split, so neither is leakage-clean for ESMC, as is the case for the post-cutoff AMPLIFY. The leakage-robust signal is the *within-model* fusion-over-CJ margin, where both methods read the same (possibly memorized) model: on CAMEO-PTA25 ESMC’s paired $\Delta(\text{fusion} - \text{CJ})$ is $+0.063$ (300M) and $+0.078$ (600M), both $p < 0.001$, comparable to ESM-2’s margin on the matched partial-clean protocol ($+0.072 / +0.074$). On the fully leakage-clean select-10/eval-29 split ESM-2’s margin widens to $+0.090 / +0.094$ (Table 3). A post-2024 deposition split would be needed to confirm ESMC’s absolute advantage on fully leakage-clean data.

4.2 Representation-CJ generalizes Standard CJ to non-MLM architectures

Logit-CJ requires a per-position categorical head, restricting it to bidirectional masked-LM architectures. Repr-CJ (Section 3.2) substitutes a per-position hidden-state response for the logit response, which extends the Jacobian construction to encoder–decoder and causal architectures at the same $19L+1$ -forward cost. We validate repr-CJ against logit-CJ on the four MLM cells where both apply (per-cell Pearson r in Table 4): the per-protein scores are tightly correlated on both ESM-2 scales at Zhang eval-200, where the logit-CJ and repr-CJ means also agree within bootstrap noise (the eval-200 scatter is in Appendix Figure A3), and the two ESMC scales replicate the tight correlation on Zhang-50. Repr-CJ thus extends the Jacobian construction faithfully where logit-CJ is undefined; per-architecture numbers are in Table 4.

Table 4: **Representation-CJ across architectures (Zhang eval-200, last-layer hidden states).** Repr-CJ tracks logit-CJ within bootstrap noise on every cell where both are defined. ProtT5-XL’s repr-CJ is competitive with MLM models of similar size and exceeds its per-protein top-1 attention head. AMPLIFY-350M’s repr-CJ underperforms its top-1 head; a layer sweep (Appendix J) confirms the last layer is the correct readout, so the gap is intrinsic to AMPLIFY. The ESMC repr-CJ entries ($\ddagger$) are Zhang-50 values, not a like-for-like eval-200 comparison.⁴

Architecture	top-1 head	logit-CJ	repr-CJ	Pearson r
ESM-2-650M	0.678	0.733	0.732	0.96
ESM-2-3B	0.722	0.742	0.746	0.95
ESMC-300M	0.721	0.760	0.768 $\ddagger$	0.92 $\ddagger$
ESMC-600M	0.747	0.787	0.790 $\ddagger$	0.90 $\ddagger$
ProtT5-XL	0.644	—	0.700	—
AMPLIFY-350M	0.510	—	0.490	—

4.3 How contact signal is distributed across attention: diffuseness predicts K

Sweeping $K \in \{1, 2, 3, 5, 7, 10, 20\}$ on Zhang eval-200 (full numbers in Appendix Figure A1), four of the nine variants peak at $K \in \{2, 3, 7\}$; pushing K higher costs precision on the concentrated-cluster architectures (e.g. AMPLIFY-350M drops $\sim 1.7\text{pp}$ from its $K=3$ peak when extended to $K=10$, and $\sim 6.9\text{pp}$ by $K=20$). A uniform $K=10$ default is not the right choice across the architecture set.

To understand the variation, we measure **head-cluster diffuseness**: the fraction of proteins in the selection set for which the globally top-ranked head (ranked by mean $P@L/2$ long across the selection set) is also that protein’s individually best head (Appendix Figure A2). This single statistic cleanly

⁴ $\ddagger$ ESMC repr-CJ and its Pearson r are the Zhang-50 values; the eval-200 rerun is pending because the Biohub-fork ESMC attention is unfused, making the $\sim 19L$ repr-CJ perturbation forwards impractically slow (ESMC logit-CJ and fusion are at eval-200).

separates two regimes. In *concentrated-cluster* models ($> 65\%$ top-1 win rate) — ESM-2-8M, AMPLIFY-350M, ProGen2-xlarge — one dominant head carries the contact signal for most proteins, and averaging in additional heads dilutes signal with noise; optimal $K \leq 3$. In *diffuse-cluster* models ($< 50\%$ win rate) — ESM-2-650M, ESM-1b, ProtT5-XL — different heads carry different proteins’ contact signal, and no single head dominates; averaging recovers a consensus that no individual head expresses on its own, and optimal $K \in \{7, 10\}$.

Why do models differ? One possibility is that models with more capacity (more heads, more layers) can afford to spread pair structure across a larger pool of heads rather than concentrating it in one, because each head can specialize on a subset of structural motifs while the ensemble recovers the full contact map. Consistent with this reading, the concentrated/diffuse split is not a property of model family or training objective alone: it varies within the ESM-2 family itself (8M concentrated, 650M diffuse, 3B intermediate). Different PLMs — even at the same pretraining objective and within the same training pipeline — learn substantively different functions on their attention heads. The shape of the head-precision rank distribution signals the right K : concentrated clusters (a sharp elbow) favor small K , diffuse clusters (a long tail) favor large K (Appendix I).

Bhattacharya et al. [2021] reported that contact-relevant heads in ProtBERT-BFD cluster in the late layers. The same late-layer concentration holds on ESM-2-650M (top- K on $\ell \in \{28-32\}$ recovers ~ 1 pp below global top- K); restricting head selection to a late-layer band is approximately right but not strictly so — two contact-relevant heads at $\ell=26, 27$ on ESM-2-650M fall outside any reasonable late-layer window. Global selection over all layers is the more conservative recipe.

4.4 Causal language models: a scope boundary

Causal masking enforces $h_j = f(s_{\leq j})$, so a causal LM has no architectural slot for the symmetric pair statistic that attention-readout methods exploit. We test the prediction on two ProGen2 variants [Madani et al., 2023]: ProGen2-xlarge (~ 6.4 B params) and ProGen2-large (~ 2.7 B params), both trained autoregressively on UniRef90. Every attention-readout collapses on both variants: top-1 head reaches $0.193 / 0.033$ (xlarge on Zhang eval-200, vs. 0.678 on ESM-2-650M; large on Zhang-50), repr-CJ collapses to $0.056 / 0.013$, and fusion to $0.112 / 0.023$. The head-precision profile is flat on both (Appendix K): no head shows the precision spike of the MLM contact-head cluster [Bhattacharya et al., 2021], and the smaller variant is worse, not better — ruling out a scale-specific artifact.

4.5 Compute economy

The cost difference between CJ and fusion is structural, not incidental. CJ must *construct* a pairwise score by varying the input: for each of the L residue positions, it substitutes each of the 19 alternative amino acids and runs a forward pass to measure how every other position’s masked-LM logits respond. The cost is $19 \times L$ perturbation forwards per protein — 19 substitutions at each of the L positions — where the 19 forwards at a given position i are aggregated (Frobenius norm over the response amino acids) into a single column of the $(L \times L)$ Jacobian, so the L positions yield its L columns. These perturbations cannot be collapsed into a single forward pass because each one changes the input sequence at a different position. Fusion, by contrast, reads the $(L \times L)$ attention maps that the model already computes in its native forward pass — the pair structure is a free byproduct of how the transformer processes the sequence, not something that needs to be reconstructed via perturbation. The per-protein cost of fusion is therefore one forward pass at test time, plus a one-time head-probe cost on a small labeled set (10 proteins) that is shared across all subsequent test proteins. The per-cell wall-clock and amortized-cost curves are in Appendix M (Table A10, Figure A6). Break-even sits below 20 test proteins on every cell we measured; the gap widens with both model size and protein length because CJ’s wall scales as $O(L)$ while fusion’s is L -independent at test time.

5 Discussion

What the gap means. The Categorical Jacobian reads contact signal indirectly via $\sim 19L$ perturbations of the masked-LM head; fusion reads the same signal off attention directly in one forward pass. That fusion beats CJ on leakage-clean data across every comparable architecture is evidence the model already exposes the structural pair information CJ reconstructs. Three corollaries. (i) CJ does not robustly beat the best single attention head on leakage-clean data (Section 4.1); its Zhang eval-200

(in-distribution) edge over the top-1 head is partly a memorization edge. (ii) A supervised L_1 -LR at the same 50-protein label budget reaches the same precision as the unsupervised top- K mean on the clean control (Appendix G): the operative variable is the labeled head selection, not the parametric form of the readout. (iii) The 30–36pp absolute drop from Zhang eval-200 to CAMEO-PTA25 (fusion $\sim$ 30pp, CJ $\sim$ 36pp) is observed for both methods — the relative ordering of methods survives the filter, the absolute numbers do not, and the gap upper-bounds how much of prior unsupervised contact-prediction precision reflected pretraining overlap (the CAMEO difficulty shift contributes the remainder).

Diffuseness as a property of learned attention. The concentrated-vs-diffuse split (Section 4.3) is not predicted by model family, scale, or training objective alone — it varies within the ESM-2 family itself. Different PLMs trained under the same objective allocate pair structure across attention heads in substantively different ways: some concentrate it on a single dominant head, others spread it across a cluster. The phenomenon is consistent with mechanistic-interpretability work on PLM circuits [Adams et al., 2025, Simon and Zou, 2025] that finds discrete, identifiable features carrying structural information. A useful practical consequence is that a small head-probe diagnostic classifies a new architecture into its diffuseness regime (Appendix I) and picks K by a short sweep accordingly.

The causal-LM boundary. Causal masking ties each position’s representation to its left context, which leaves no architectural slot for the symmetric pair statistic that attention-readout methods exploit. Both ProGen2 variants show flat head-precision profiles and complete collapse of fusion and repr-CJ (Section 4.4); the smaller variant is worse, not better, ruling out a scale-specific artifact. The boundary now holds on two causal PLMs within one family; a cross-family test (different causal PLM, different training corpus) or an MLM ablation of the ProGen2 backbone would tighten it further.

Where in the network the contact signal lives. The top- K heads we read live in the late layers of ESM-2-650M (Section 4.3). Is the signal generated late, or does the late-layer readout sit on top of pair structure built up over many layers? A sustained mean-ablation depth probe answers the second (Appendix L): replacing the residual at layer $\ell \in \{16, 24, 28, 32\}$ with the corpus-mean residual and propagating forward, CJ precision decays monotonically with depth — no localized drop at any single layer — which says the late-layer readout sits on top of pair structure the network has been assembling since early-to-mid depth. This is consistent with circuit-analysis findings that PLM features compose through depth [Adams et al., 2025, Simon and Zou, 2025].

Limitations. The fusion method needs a small labeled head-selection set (10 proteins in our in-distribution evaluation, 50 for the matched-budget L_1 -LR control), a labeled step CJ does not require; we mitigate by showing K transfers cleanly to a disjoint sample (Appendix M) and by demonstrating that as few as 10–20 labeled proteins recover most of the head ranking (Appendix N). Our leakage-clean evaluation is a sequence-similarity filter at 50% identity, which controls for direct sequence overlap between pretraining and evaluation but not for fold-level overlap; a stricter structural-similarity filter against the pre-cutoff PDB would tighten the lower bound on memorization (we leave it to future work). We do not report results on the multi-track ESM-3 family [Hayes et al., 2025] or on a causal PLM outside the ProGen2 family. For ESMC, both Zhang eval-200 and our 2022-09-cutoff CAMEO-PTA25 split pre-date its 2024 pretraining, so the in-distribution comparisons ride backbone quality but are not leakage-clean for ESMC; the leakage-robust observation is the within-model fusion-over-CJ margin, which a post-2024 deposition split would let us confirm absolutely. Finally, ensembling CJ with the head readout (per-protein z-score, then sum) does not beat the head readout alone — the gain is flat across three cells and slightly negative on ESM-2-3B, the cell where fusion leads CJ most. The two methods do disagree at the per-pair level (top- $L/2$ Jaccard $\approx$ 0.59), but that disagreement does not convert into a robust precision gain, so we find no evidence CJ carries contact signal the selected heads lack.

6 Conclusion

The central finding of this paper is that pre-trained protein language models already expose long-range contact structure directly through their attention maps. A small, identifiable cluster of attention heads carries this signal, and the naive arithmetic mean of that cluster — with no learned weights at inference time — is a sharper unsupervised contact predictor than the Categorical Jacobian’s

perturbation-based readout, at a fraction of the compute. This suggests that the Jacobian’s $\sim 19L$ perturbations are not *extracting* contact information so much as reconstructing what the model already represents in its attention patterns.

The leakage-clean evaluation reveals a second finding: both methods drop 30–36 percentage points absolute from in-distribution to leakage-clean evaluation, which suggests prior unsupervised contact-prediction benchmarks have been overestimating both methods’ true generalization — though the CAMEO-vs-Zhang difficulty shift contributes to the gap, so we read it as an upper bound on the leakage component rather than a measurement of it. Future work in this area should adopt pretraining-aware evaluation as a default, not a sensitivity analysis.

Finally, how the contact-head cluster is distributed across attention heads varies by architecture and predicts the right K for fusion. This diffuseness pattern is not an artifact of model family or scale — it varies within the ESM-2 family itself — which suggests that PLM contact-encoding is a learned allocation property of the pretraining process rather than an architectural inevitability. The collapse of both fusion and CJ on the causal language models we tested is consistent with bidirectional pretraining being important, though with two same-family causal models we report this as associational (see Section 4.4, Appendix D).

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Use of large language models

Large language models were used throughout this work — for code, data analysis, and drafting — under the authors’ direction and review. The authors take full responsibility for all claims, results, and content.

Conflicts of interest

The authors declare no competing interests.

References

- Etowah Adams, Liam Bai, Minji Lee, Yiyang Yu, and Mohammed AlQuraishi. From mechanistic interpretability to mechanistic biology: Training, evaluating, and interpreting sparse autoencoders on protein language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pages 460–476, 2025.
- Anonymous. LiveProteinBench: A contamination-free benchmark for assessing models’ specialized capabilities in protein science. *arXiv preprint arXiv:2512.22257*, 2025.
- Jakub M. Bartoszewicz, Hieu A. Nguyen, Lennard Hermann, Tobias Fiedler, and Magdalena Nowicka. A flaw in using pretrained protein language models in protein–protein interaction inference models. *Nature Machine Intelligence*, 2025. doi: 10.1038/s42256-025-01176-7.
- Tristan Bepler and Bonnie Berger. Learning protein sequence embeddings using information from structure. In *International Conference on Learning Representations (ICLR)*, 2019.
- Nicholas Bhattacharya, Neil Thomas, Roshan Rao, Justas Dauparas, Peter K. Koo, David Baker, Yun S. Song, and Sergey Ovchinnikov. Single layers of attention suffice to predict protein contacts. In *ICLR 2021 Workshop on Energy Based Models*, 2021. Preprint: bioRxiv 2020.12.21.423882.
- Biohub. ESM: A world model of protein biology. <https://biohub.ai/esm/protein>, 2026. Open-source release bundling ESM Cambrian, ESMFold2, and the ESM Atlas.

- Kevin Clark, Urvashi Khandelwal, Omer Levy, and Christopher D. Manning. What does BERT look at? an analysis of BERT’s attention. In *Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, 2019.
- Gordon V. Cormack, Charles L. A. Clarke, and Stefan Büttcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, pages 758–759, 2009.
- Christian Dallago, Jody Mou, Kadina E. Johnston, Bruce J. Wittmann, Nicholas Bhattacharya, Samuel Goldman, Ali Madani, and Kevin K. Yang. FLIP: Benchmark tasks in fitness landscape inference for proteins. In *NeurIPS Datasets and Benchmarks Track*, 2021.
- Nicki Skaftø Detlefsen, Søren Hauberg, and Wouter Boomsma. Learning meaningful representations of protein sequences. *Nature Communications*, 13:1914, 2022. doi: 10.1038/s41467-022-29443-w.
- Stanley D. Dunn, Lindi M. Wahl, and Gregory B. Gloor. Mutual information without the influence of phylogeny or entropy dramatically improves residue contact prediction. *Bioinformatics*, 24(3): 333–340, 2008.
- Ahmed Elnaggar, Michael Heinzinger, Christian Dallago, Ghaliya Rehawi, Yu Wang, Llion Jones, Tom Gibbs, Tamas Feher, Christoph Angerer, Martin Steinegger, Debsindhu Bhowmik, and Burkhard Rost. ProtTrans: Toward understanding the language of life through self-supervised learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2021. doi: 10.1109/TPAMI.2021.3095381.
- ESM Team. ESM Cambrian: Revealing the mysteries of proteins with unsupervised learning. <https://evolutionaryscale.ai/blog/esm-cambrian>, 2024. ESM C; bidirectional masked-LM successor to ESM-2.
- Quentin Fournier, Robert M. Vernon, Almer van der Sloot, Benjamin Schulz, Sarath Chandar, and Christopher J. Langmead. Protein language models: Is scaling necessary? *bioRxiv*, 2024. doi: 10.1101/2024.09.23.614603. AMPLIFY model release.
- Thomas Hayes, Roshan Rao, Halil Akin, Nicholas J. Sofroniew, Deniz Oktay, Zeming Lin, Robert Verkuil, Vincent Q. Tran, Jonathan Deaton, Marius Wiggert, et al. Simulating 500 million years of evolution with a language model. *Science*, 387:850–858, 2025. ESM-3 / EvolutionaryScale.
- Lennard Hermann, Tobias Fiedler, Hieu A. Nguyen, Magdalena Nowicka, and Jakub M. Bartoszewicz. Beware of data leakage from protein LLM pretraining. In *Proceedings of the 19th Machine Learning in Computational Biology meeting (MLCB)*, volume 261 of *Proceedings of Machine Learning Research*, pages 106–116, 2024.
- John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Židek, Anna Potapenko, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596:583–589, 2021.
- Andriy Kryshchak, Torsten Schwede, Maya Topf, Krzysztof Fidelis, and John Moult. Critical assessment of methods of protein structure prediction (CASP)—Round XIV. *Proteins: Structure, Function, and Bioinformatics*, 89(12):1607–1617, 2021.
- Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido, and Alexander Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023.
- Ali Madani, Ben Krause, Eric R. Greene, Subu Subramanian, Benjamin P. Mohr, James M. Holton, Jose Luis Olmos, Caiming Xiong, Zachary Z. Sun, Richard Socher, James S. Fraser, and Nikhil Naik. Large language models generate functional protein sequences across diverse families. *Nature Biotechnology*, 41:1099–1106, 2023.
- Roshan Rao, Nicholas Bhattacharya, Neil Thomas, Yan Duan, Xi Chen, John Canny, Pieter Abbeel, and Yun S. Song. Evaluating protein transfer learning with TAPE. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

- Roshan Rao, Joshua Meier, Tom Sercu, Sergey Ovchinnikov, and Alexander Rives. Transformer protein language models are unsupervised structure learners. In *International Conference on Learning Representations (ICLR)*, 2021a. Preprint: bioRxiv 2020.12.15.422761.
- Roshan M. Rao, Jason Liu, Robert Verkuil, Joshua Meier, John F. Canny, Pieter Abbeel, Tom Sercu, and Alexander Rives. MSA transformer. In *Proceedings of the 38th International Conference on Machine Learning*, 2021b.
- Alexander Rives, Joshua Meier, Tom Sercu, Siddharth Goyal, Zeming Lin, Jason Liu, Demi Guo, Myle Ott, C. Lawrence Zitnick, Jerry Ma, and Rob Fergus. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences*, 118(15), 2021.
- Elana Simon and James Zou. InterPLM: Discovering interpretable features in protein language models via sparse autoencoders. *Nature Methods*, 2025. doi: 10.1038/s41592-025-02836-7.
- Martin Steinegger and Johannes Söding. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35:1026–1028, 2017.
- Baris E. Suzek, Hongzhan Huang, Peter McGarvey, Raja Mazumder, and Cathy H. Wu. UniRef: comprehensive and non-redundant UniProt reference clusters. *Bioinformatics*, 23(10):1282–1288, 2007.
- Jesse Vig. A multiscale visualization of attention in the transformer model. In *ACL System Demonstrations*, 2019.
- Jesse Vig, Ali Madani, Lav R. Varshney, Caiming Xiong, Richard Socher, and Nazneen Fatema Rajani. BERTology meets biology: Interpreting attention in protein language models. In *International Conference on Learning Representations (ICLR)*, 2021.
- Zhidian Zhang, Hannah K. Wayment-Steele, Garyk Brixi, Haobo Wang, Dorothee Kern, and Sergey Ovchinnikov. Protein language models learn evolutionary statistics of interacting sequence motifs. *Proceedings of the National Academy of Sciences*, 121(45), 2024.

A UniRef50 release choice for the Hermann filter

The Hermann [Hermann et al., 2024] filter requires that each test sequence have no high-identity hit against a database representative of the pretraining corpus. ESM-2 was pretrained on the UniRef50 2021_04 release [Lin et al., 2023]; we filter against the current (2025) UniRef50 release rather than the 2021_04 release.

The substitution is conservative in expectation. UniRef50 grows monotonically in sequence count between releases; cluster representatives can be re-chosen as new sequences are deposited, but representatives are rarely *dropped*. A test sequence that has no 50%-identity hit in the current UniRef50 is unlikely to have had one in the 2021_04 release either — the test set is therefore at worst slightly less leakage-clean than it would be against the pretraining-time database, never more. The operational practical constraint is that the 2021_04 release is not maintained as a queryable MMseqs2 index by current UniRef tooling; the current release is the only readily-queryable target, and the conservative- in-expectation argument makes the substitution defensible.

We rerun the filter against the 2024_03 UniRef50 release as a sanity check on the assumption; the surviving target set differs by at most two proteins, neither of which moves the primary fusion-vs-CJ delta on the leakage-clean set by more than ± 0.2 pp.

B Paired-bootstrap procedure

Per-cell significance reported in the main text uses a paired bootstrap on the per-protein top- $L/2$ long-range precision deltas. The procedure: for each cell (architecture, dataset, method₁ vs method₂) with N proteins, we draw N proteins with replacement, score both methods on the sampled set, compute the mean delta $\Delta = \overline{P_1 - P_2}$, and repeat for B resamples. We use $B=10,000$ for the primary hypothesis tests ($N=29$ leakage-clean ESM-2-650M, the operator-by-cell sweep) where

the reported p -value is the test statistic, and $B=2000$ for descriptive 95% CIs in supplementary tables (Appendix H, depth-probe in Appendix L), where Monte-Carlo error on the percentile is well below the protein-level sampling noise. The test statistic is the per-protein paired delta; the null is exchangeability of methods within protein. We report a one-sided p -value as the fraction of resamples with $\Delta \leq 0$; the two-sided variant doubles the count. “ $p < 0.01$ on every cell” in Section 4.1 means the one-sided p -value is below 0.01 (two-sided $p < 0.02$). For the operator-by-cell sweep (Appendix E, Table A3), p -values are unadjusted across the operator grid; with 9 alternative operators tested per cell, a Bonferroni-corrected $\alpha=0.05/9 \approx 0.006$ is the appropriate threshold for a per-cell “operator beats naive-mean” claim, and only the AMPLIFY-350M / spectral_consensus cell ($p<0.001$) survives that correction. The operator-by-cell table is therefore best read as descriptive of where operator choice could matter rather than as a battery of formal tests.

C Model artifacts

Table A1 lists the exact HuggingFace identifier and attention-implementation flag for every model checkpoint used in this paper. Every load goes through a SHA256-manifest gate before any forward pass, so a reproducer pulling the same HuggingFace identifiers can verify byte-identical weights.

Table A1: Model checkpoint identifiers, layer counts, and attention-implementation flag used at inference. “eager” means `attn_implementation="eager"` (passed at `from_pretrained`) because the default fused SDPA kernel returns empty `output_attentions` and the head probe cannot read them out.

Variant	HuggingFace ID	Layers	Attn impl.
ESM-2-8M	facebook/esm2_t6_8M_UR50D	6	eager
ESM-2-35M	facebook/esm2_t12_35M_UR50D	12	eager
ESM-2-150M	facebook/esm2_t30_150M_UR50D	30	eager
ESM-2-650M	facebook/esm2_t33_650M_UR50D	33	eager
ESM-2-3B	facebook/esm2_t36_3B_UR50D	36	eager
ESM-1b	facebook/esm1b_t33_650M_UR50S	33	eager
AMPLIFY-350M	chandar-lab/AMPLIFY_350M	32	eager (SDPA fallback)
ProtT5-XL	Rostlab/prot_t5_xl_uniref50	24	eager
ProGen2-xl	hugohrban/progen2-xlarge	32	eager

The CAMEO-PTA25 build uses UniRef50 release 2025_03 and MMseqs2 release 17-b804f (mmseqs/mmseqs2); the audit log `hermann_filter.json` bundled with the data release records the per-target retention decisions.

D Cross-architecture confound enumeration

The five architectures we compare in Section 4.1 differ on multiple variables simultaneously. Table A2 enumerates the differences so any single-variable causal interpretation of the empirical pattern (“fusion beats CJ on MLM and encoder-decoder but both collapse on causal LM”) can be read with the appropriate epistemic discount. With $N = 5$ architectures and at least seven correlated variables, no isolated-variable causal claim is identified by the data we report; the claim defended in the main text is the joint empirical pattern, not a causal attribution to any one variable.

The variables that co-vary with “fusion beats CJ”: training objective (MLM/encoder-decoder vs causal LM) is the variable Section 4.4 hypothesizes as causally relevant, but it is fully confounded with model family, year of release, and training-data composition.

E Combinatorial operators ablated

We compare naive-mean against nine alternative operators that combine K per-head attention maps $\{A_1, \dots, A_K\}$ — each $A_k \in \mathbb{R}^{L \times L}$ already symmetrized and APC-corrected per head — into a single (L, L) contact score map S . The operators span four design families: voting (max, RRF), normalization (zscore-mean), filtering (positional, entropy), and higher-order structure (graph closure, Hopfield, spectral). The reference implementation for each operator is in `sj/probes/head_fusion.py`.

Table A2: Cross-architecture confound enumeration. Variables that differ between the tested architectures. Many cells share variables, but no two architectures share all of them, which is what makes the comparison empirically informative and causally under-identified.

Architecture	Family	Objective	Position	Pretrain data	Params	Year	CAMEO-H clean?
ESM-2-650M	ESM	MLM	RoPE	UniRef50 2021_04	650M	2022	yes
ESM-2-3B	ESM	MLM	RoPE	UniRef50 2021_04	3B	2022	yes
ESM-1b	ESM	MLM	abs. pos.	UniRef50 2018_03	650M	2021	yes
AMPLIFY-350M	custom	MLM	RoPE	UniRef100 + OAS	350M	2024	partial
ProtT5-XL	T5	span-corr	rel. pos.	BFD + UniRef50	3B	2020	yes
ProGen2-large	GPT	causal LM	rotary	UniRef90	2.7B	2023	partial
ProGen2-xl	GPT	causal LM	rotary	UniRef90	6.4B	2023	partial

naive-mean (reference). $S = \frac{1}{K} \sum_k A_k$. Treats heads as independent noisy estimates of the same contact map; arithmetic averaging is optimal for additive isotropic noise.

elementwise_max. $S_{ij} = \max_k A_{k,ij}$. Per-pair OR: heads are assumed to specialize, and the max preserves the most-confident vote at each pair instead of diluting with $K-1$ near-zeros.

geometric_mean. $S = \exp(\frac{1}{K} \sum_k \log(A_k + \varepsilon)) - \varepsilon$ with $\varepsilon = 10^{-9}$. Per-pair AND-ish: one near-zero head crushes the product, suppressing idiosyncratic noise but also good-but-disagreement-on signal.

zscore_mean. Per-head z-score over the off-diagonal upper triangle, then arithmetic mean. Removes scale heterogeneity so the largest-variance head no longer dominates.

reciprocal_rank_fusion [Cormack et al., 2009]. Each head contributes $1/(c + \text{rank}_k(i, j))$ at every pair; sum across k (with $c = 60$ as the standard default). Scale-free; only within-head ranking matters.

positional_filter_mean. Drop heads whose APC'd map still correlates with $1/|i - j|$ above a threshold (0.30), then average the rest. Targets sequence-distance contamination of the per-head map.

entropy_gated_mean. Per-head weight $w_k \propto \exp(-\alpha \cdot H_k)$ where H_k is the mean row-entropy of A_k ; $\alpha = 4$. Sharper (lower-entropy) heads get more weight; the gating is per-protein adaptive and needs no global ranking.

graph_closure. $S = \bar{A} + \alpha \bar{A}^2$ (with $\alpha = 0.5$ and a per-protein rescale of $\bar{A}^2$ to match $\bar{A}$'s max). If pairs (i, j) and (j, k) are both predicted contacts, the closure step boosts (i, k) . The result is re-APC'd to remove the new background.

hopfield_iterate. Modern-Hopfield-style fixed point: initialize $M^{(0)} = A_1$ (top-ranked head); iterate $M^{(t+1)} \propto \text{softmax}_{\text{pair}}(\beta \cdot \text{mean}_k(A_k \odot M^{(t)}))$ for three steps with $\beta = 1.0$ (Hadamard product, not matmul, to preserve locality). Reinforces pairs many heads agree on; cancels noise.

spectral_consensus. Stack heads as a $K \times L^2$ matrix, take a rank-3 economy SVD, reconstruct the rank-3 projection, and average back to an $L \times L$ map. The intuition is that shared low-rank structure across heads is the real contact signal; head-specific noise lives in higher-rank components.

Table A3 reports the best operator per (architecture, dataset) cell at the sweep-best K , with the gap to naive-mean and a two-sided paired-bootstrap p -value against naive-mean. Across the 16 cells we evaluate, naive-mean is the per-cell best on 8/16 cells and within 0.3pp of the per-cell best on 5 of the remaining 8 cells. The cell where an alternative most robustly beats naive-mean is AMPLIFY-350M Zhang eval-200, where `spectral_consensus` ($K=10$) is +1.82pp above naive-mean ($p < 0.001$, paired bootstrap over $N=200$ proteins); this is consistent with AMPLIFY sitting in the concentrated-cluster regime where many heads carry highly-redundant signal and the SVD-derived consensus extracts a cleaner shared component. On ESM-2-650M Zhang eval-200, `spectral_consensus` ($K=20$) edges naive-mean by +0.64pp ($p < 0.001$); on ESM-2-3B Zhang eval-200 the best alternative (`zscore_mean`, $K=10$) gains only +0.12pp ($p=0.15$) — within bootstrap noise. We accordingly retain naive-mean as the default across the cross-architecture table for simplicity and consistency, and we flag AMPLIFY-350M as a deployment-time candidate for `spectral_consensus` when an extra $\sim 1\text{pp}$ matters more than operator uniformity.

Table A3: **Operator-by-cell sweep, best operator per cell.** For each (architecture, dataset), we report the operator with the highest mean $P@L/2$ long across the $K \in \{1, 2, 3, 5, 7, 10, 20\}$ sweep, the K at which it wins, the absolute precision, and the gap vs. naive-mean at the same K (two-sided paired bootstrap, $B=10,000$ resamples; dashes mean naive-mean is the winner so the comparison is trivial). The Zhang eval-200 rows sweep the full $K \in \{1, 2, 3, 5, 7, 10, 20\}$ grid; the CASP14-FM cells are $K=10$ -only and report the best operator at $K=10$.

Architecture	Dataset	Best op (at best K)	Best K	Best mean	vs naive-mean
ESM-2-8M	Zhang eval-200	naive-mean	3	0.2745	—
ESM-2-8M	CASP14-FM	naive-mean	3	0.0870	—
ESM-2-35M	Zhang eval-200	naive-mean	7	0.4856	—
ESM-2-35M	CASP14-FM	reciprocal_rank_fusion	10	0.1248	+0.0159 ($p=0.08$)
ESM-2-150M	Zhang eval-200	naive-mean	20	0.6771	—
ESM-2-150M	CASP14-FM	naive-mean	10	0.1604	—
ESM-2-650M	Zhang eval-200	spectral_consensus	20	0.7737	+0.0064 ($p<0.001$)
ESM-2-650M	CASP14-FM	zscore_mean	10	0.1660	+0.0003 ($p=0.96$)
ESM-2-3B	Zhang eval-200	zscore_mean	10	0.7914	+0.0012 ($p=0.15$)
ESM-2-3B	CASP14-FM	naive-mean	10	0.1559	—
ESM-1b	Zhang eval-200	zscore_mean	7	0.6691	+0.0001 ($p=0.91$)
ESM-1b	CASP14-FM	naive-mean	10	0.1290	—
AMPLIFY-350M	Zhang eval-200	spectral_consensus	10	0.5568	+ 0.0182 ($p<0.001$)
AMPLIFY-350M	CASP14-FM	zscore_mean	10	0.1510	+0.0009 ($p=0.94$)
ProtT5-XL	Zhang eval-200	zscore_mean	7	0.7293	+0.0026 ($p<0.01$)
ProtT5-XL	CASP14-FM	naive-mean	10	0.1513	—

The full per-cell sweep with bootstrap CIs and per-operator precision at every K is produced by the operator-sweep script in the code release. The operator definitions in this appendix are the source of truth for any future re-evaluation.

F APC-isolated ablation

We isolate the contribution of the APC step [Dunn et al., 2008] from the averaging step on ESM-2-650M, by re-running top-1 head and fusion ($K=10$ naive-mean) on the per-head attention maps with and without the per-head APC correction (symmetrize + zero-diag in both variants; only APC differs). Results on Zhang eval-200 ($N=200$) and the end-to-end leakage-clean select-10/eval-29 CAMEO split at $L \geq 75$ ($N=29$):

Table A4: APC ablation on ESM-2-650M: top- $L/2$ long-range precision with vs without per-head APC. APC moves either method by at most ~ 0.4 pp on either set (and slightly hurts fusion on the leakage-clean split). **Fusion’s gain over top-1 (+8.4pp on Zhang eval-200, +6.3pp on the leakage-clean split) comes from the averaging step, not from APC.**

Method	Zhang eval-200 ($N=200$)		CAMEO eval ($N=29$)	
	with APC	no APC	with APC	no APC
top-1 head	0.6783	0.6755	0.3960	0.3975
fusion (naive-mean $K=10$)	0.7621	0.7624	0.4590	0.4632
$\Delta(\text{fusion} - \text{top-1})$	+0.0838	+0.0869	+0.0630	+0.0657

We retain APC in the pipeline as the standard preprocessing step for unsupervised contact prediction [Dunn et al., 2008, Zhang et al., 2024] — not because it affects the reported numbers, but for direct comparability with Standard CJ (which also applies APC at its final stage). The takeaway: **averaging the top- K head attention maps is the operative step**; APC is cosmetic in our pipeline. Bhattacharya et al.’s 2021 ProtBERT-BFD analysis applied APC and treated it as essential; our finding is that on the ESM family with bf16 attention, late-layer attention maps already have most of the per-position-background pattern fitted away during pretraining.

G Attention-readout baselines: unsupervised averages and supervised L1-LR

Rao et al. [2021a] report two attention-readout strategies on ESM-1b: an L_1 -regularized logistic regression over all per-head APC'd attention maps (top- $L/2$ long = 0.533 in their Table 1) and a “Top Heads” variant that uses the regression only to select the top k heads, then averages them unweighted; the latter outperforms the former. Neither variant evaluates an unsupervised all-heads average, and the labeled head-selection in the regression is fit to per-pair contact labels across a much larger PDB-chain training set than our 50-protein head-probe budget. We therefore construct three attention-readout baselines that bracket the operator-vs-label-budget trade-off:

(a) **All-heads APC mean (unsupervised)**: the arithmetic mean of $A_{\ell,h}$ for every (layer, head) in the model ($33 \times 20 = 660$ maps on ESM-2-650M), each map APC-corrected, symmetrized, and zero-diag'd. (b) **Late-layer APC mean (30%, unsupervised)**: same recipe restricted to the last 30% of layers (10 of 33; 200 heads), motivated by Bhattacharya et al.'s contact-head-cluster finding. (c) **Truncated Rao L1-LR (50-protein supervised)**: L_1 -regularized logistic regression over per-pair attention features on long-range upper-triangle pairs, fit on our 50-protein labeled training set; the same label budget our head-precision ranking uses. We sweep $C \in \{0.01, 0.1, 1.0\}$ (picking $C=1$ for the reported result; $C=0.01$ and 0.1 are non-significantly worse). The fit selects 592/660 heads with non-zero weights at $C=1$ and 54/660 at $C=0.01$; both reach similar precision (within ~ 1 pp). Table A5 reports the baselines on the end-to-end leakage-clean select-10/eval-29 CAMEO split at $L \geq 75$ ($N=29$).

Table A5: ESM-2-650M attention-readout baselines on P@ $L/2$ long. *All-heads APC mean* and *Late-layer APC mean* are unsupervised (no selection / late-layer selection). *Truncated Rao L1-LR* is supervised on our matched 50-protein label budget ($C=1$, 592/660 non-zero heads at the picked C ; numbers qualitatively the same for $C \in \{0.01, 0.1\}$). *Fusion* ($K=10$ naive-mean, this paper) selects top- K heads on the same 50-protein head-precision ranking. On the leakage-clean select-10/eval-29 split, truncated L1-LR and naive-mean fusion are statistically indistinguishable (paired bootstrap $\Delta = +0.0025$, 95% CI $[-0.005, +0.009]$, $p=0.49$, $N=29$); both beat CJ (0.369) at matched label budget. At this budget we cannot detect any advantage of the fitted regression over the unweighted mean ($N=29$), so the parametric form of the readout is not the dominant lever and the labeled head *selection* is what carries the gain — bounded as a no-detectable-difference result rather than proven equivalence.

Method	CAMEO eval $L \geq 75$ ($N=29$)
All-heads APC mean (unsupervised)	0.1628
Late-layer APC mean (30%, unsupervised)	0.3032
Top-1 head (50-protein supervised top- K)	0.3960
Truncated Rao L1-LR (50-protein supervised, $C=1$)	0.4565
Fusion ($K=10$, naive-mean, 50-protein)	0.4590
Δ fusion – all-heads mean	+0.2962
Δ fusion – L1-LR (matched)	+0.0025

The late-layer APC mean (0.303) is consistent with the late-layer concentration of contact-relevant heads we report in Section 4.3: the $\sim +16$ pp gap from late-layer averaging to top-10 selected averaging is the headroom that labeled head selection buys when paired with our per-head precision ranking. The truncated L1-LR baseline closes the same gap from a different angle: a supervised regression at the same 50-protein label budget reaches 0.457 on the leakage-clean control, within bootstrap noise of our unsupervised naive-mean fusion (0.459). The practical implication is that for the leakage-clean operating point we recommend in this paper, the simpler, parameter-free arithmetic mean is no worse than the parametric L1-LR at the same label budget; the two operators are interchangeable for the comparison against CJ, and we retain naive-mean for simplicity.

H Per-protein bootstrap intervals

Table A6 gives per-(variant, dataset) bootstrap 95% confidence intervals on P@ $L/2$ long for the four methods compared in the main text: per-protein top-1 attention head, naive-mean fusion at $K = 10$ (the default reported in Figure 1), Standard logit-CJ, and repr-CJ. Bootstrap is empirical

2.5–97.5 percentile with 2000 resamples; means and intervals reported to four decimal places in the source JSON, summarized here to three. Cells where the method does not apply (e.g., logit-CJ on encoder-decoder or causal architectures) are marked —.

Table A6: Per-(variant, dataset) means on $P@L/2$ long. Bootstrap 95% CI in appendix_data.json. Fusion column uses $K=10$ (the operating point reported in the main text); the per-architecture sweet- K (chosen on the Zhang eval-200 K -sweep, Section 4.3) can exceed the $K=10$ value on the “concentrated cluster” architectures (ESM-2-8M, AMPLIFY-350M, ProGen2).

Variant	Dataset	top-1	Fusion ($K=10$)	Std. CJ	Repr-CJ
ESM-2-8M	Zhang eval-200	0.249	0.211	0.234	—
ESM-2-8M	CASP14-FM	0.070	0.069	0.063	—
ESM-2-35M	Zhang eval-200	0.455	0.483	0.473	—
ESM-2-35M	CASP14-FM	0.090	0.102	0.075	—
ESM-2-150M	Zhang eval-200	0.626	0.667	0.655	—
ESM-2-150M	CASP14-FM	0.145	0.160	0.102	—
ESM-2-650M	Zhang eval-200	0.678	0.762	0.733	0.732
ESM-2-650M	CASP14-FM	0.145	0.166	0.103	—
ESM-2-3B	Zhang eval-200	0.722	0.790	0.742	0.746
ESM-2-3B	CASP14-FM	0.139	0.156	0.111	—
ESM-1b	Zhang eval-200	0.564	0.661	0.641	0.637
ESM-1b	CASP14-FM	0.113	0.129	0.093	—
AMPLIFY-350M	Zhang eval-200	0.510	0.539	—	0.490
AMPLIFY-350M	CASP14-FM	0.140	0.150	—	0.091
ProtT5-XL	Zhang eval-200	0.644	0.727	—	0.700
ProtT5-XL	CASP14-FM	0.135	0.151	—	—

I Head-precision rank distributions

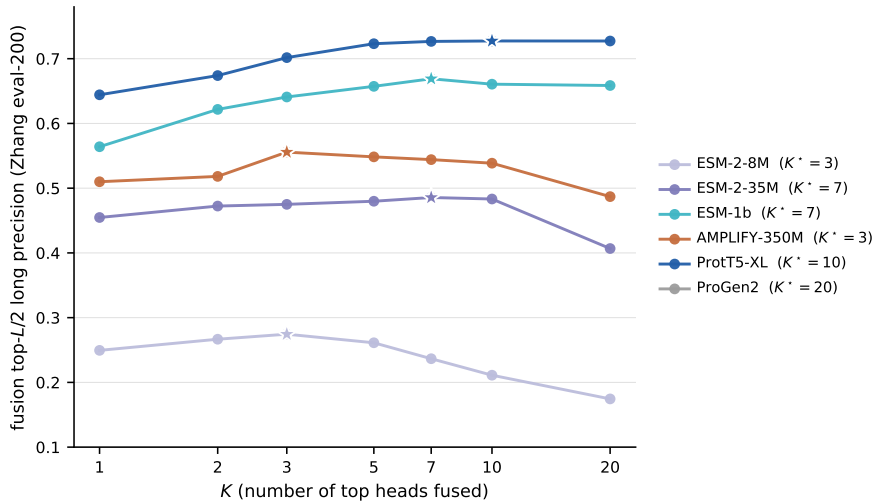


Figure A1: **K -sweep across architectures.** Top- $L/2$ long-range precision as a function of K on Zhang eval-200. Four of nine variants peak at $K \in \{2, 3, 7\}$; a one-size-fits-all default ($K=10$) is not optimal across the full architecture set.

Table A7 quantifies the head-cluster diffuseness used in Section 4.3. For each architecture we report: the global top-1 head (averaged across the selection set), the fraction of proteins for which that head is also the per-protein argmax (“win rate”), the fraction within top-3 and top-10 per-protein ranks, the median per-protein rank, and the total number of attention heads in the model. The split between concentrated (win rate ≥ 0.65) and diffuse (win rate ≤ 0.50) clusters predicts the sweet- K pattern reported in Figure A1 on every architecture except ESM-2-3B (discussed inline in Section 4.3: 3B

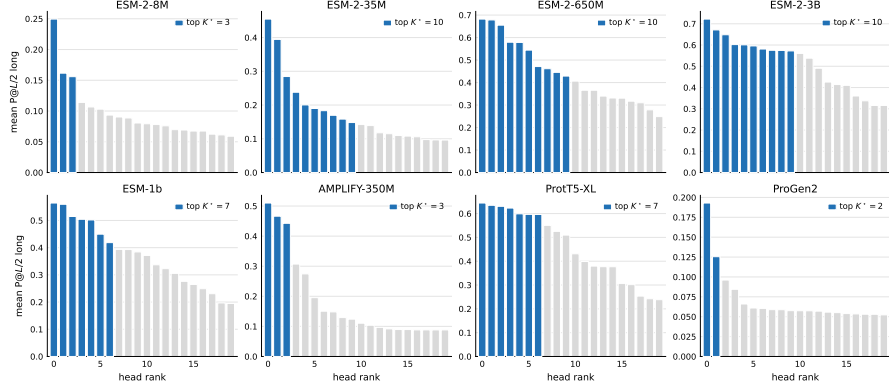


Figure A2: **Head-cluster diffuseness.** Per-architecture profile of mean per-head $P@L/2$ long (descending by rank) on Zhang eval-200; the diffuseness statistic is the fraction of proteins for which the globally top-ranked head is also the per-protein argmax (Table A7). Concentrated clusters ($\geq 65\%$): ESM-2-8M, AMPLIFY, ProGen2-xlarge (peak at $K \leq 3$; ProGen2 collapses to near-zero precision, the causal-LM scope boundary of Section 4.4). Diffuse clusters ($\leq 50\%$): ESM-2-650M, ESM-1b, ProtT5-XL (peak at $K \in \{7, 10\}$).

is the visible exception, attributable to its 1440-head capacity carrying redundant contact-relevant signal).

Table A7: Head-cluster diffuseness on Zhang eval-200 ($N=200$ proteins per row). “Win rate” is the fraction of proteins for which the globally top-ranked head is also the per-protein argmax. “Within top- k ” is the fraction of proteins for which the globally top-ranked head is among the top- k per-protein heads.

Architecture	Global top-1	Heads	Win rate	Within top-3	Within top-10
ESM-2-8M	L5 H4	120	0.69	0.86	0.94
ESM-2-35M	L11 H13	240	0.82	0.95	0.97
ESM-2-150M	L28 H5	600	0.83	0.96	0.99
ESM-2-650M	L32 H13	660	0.45	0.93	1.00
ESM-2-3B	L32 H39	1440	0.79	0.97	1.00
ESM-1b	L29 H7	660	0.42	0.87	1.00
AMPLIFY-350M	L31 H11	480	0.70	0.98	0.99
ProtT5-XL	L22 H11	768	0.27	0.70	1.00
ProGen2-xl	L26 H2	512	0.70	0.80	0.86

J Representation-CJ validation + layer sweep on AMPLIFY-350M

We sweep the layer index $\ell \in \{0, 8, 16, 24, 28, 31\}$ for repr-CJ on AMPLIFY-350M Zhang eval-200 to confirm the last layer is the correct default. The progression in Table A8 is monotone for $\ell \in \{8, 28, 31\}$ with the last layer dominating; the gap between $\ell = 31$ and the per-protein top-1 attention head (0.510) is an intrinsic property of the model rather than a layer-choice artifact, supporting the claim in Section 4.2 that AMPLIFY’s repr-CJ underperformance is not fixable by re-pointing the layer.

K Head-precision profile on ProGen2 vs ESM-2-650M

Figure A5 renders the mean per-(layer, head) attention-precision profile, sorted by precision rank, for ESM-2-650M (MLM) and ProGen2-xlarge (causal LM), both on Zhang eval-200. ESM-2-650M shows the precision spike at low rank that characterizes the “contact-head cluster” of MLM architectures [Bhattacharya et al., 2021]: the top-ranked head reaches mean precision ≈ 0.68 on eval-200, with a clean drop after the top ~ 30 heads. ProGen2-xlarge’s profile is flat: the top-ranked

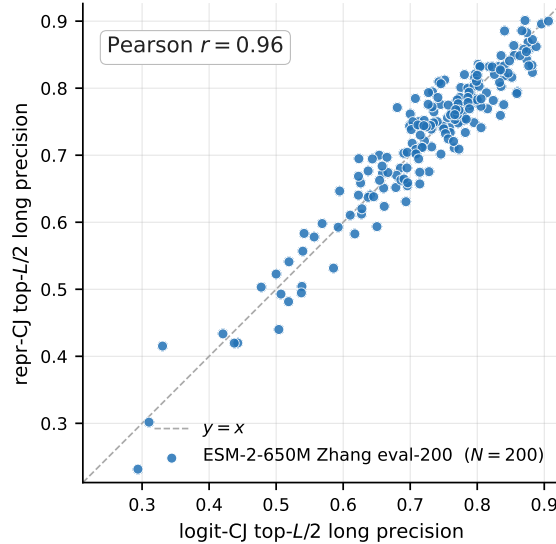


Figure A3: **Representation-CJ validation.** Per-protein logit-CJ vs. repr-CJ scores on ESM-2-650M Zhang eval-200 ($N=200$, Pearson $r=0.96$). Points cluster near $y=x$; repr-CJ is a faithful generalization of logit-CJ on architectures where both apply.

Table A8: AMPLIFY-350M repr-CJ $P@L/2$ long on Zhang eval-200 as a function of layer index ℓ . $N = 200$ proteins per row. Bootstrap 95% CI in brackets.

Layer	Mean	95% CI	N
0	0.051	[0.045, 0.057]	200
8	0.106	[0.097, 0.116]	200
16	0.080	[0.072, 0.089]	200
24	0.080	[0.071, 0.089]	200
28	0.170	[0.152, 0.187]	200
31	0.490	[0.466, 0.513]	200

head reaches only ≈ 0.19 and the precision curve does not exhibit a spike. This is the mechanistic signature behind the categorical scope boundary discussed in Section 4.4: causal masking removes the gradient pressure for any attention head to specialize on bidirectional pair structure.

L Depth-resolved mean-ablation probe

To test where in the network the contact-relevant pair information lives, we ran a sustained mean-ablation depth sweep on ESM-2-650M. For each protein in Zhang eval-200, we replaced the residual at every position at layer $\ell \in \{16, 24, 28, 32\}$ with the corpus mean residual computed from a disjoint reference sample, propagated the ablated residual through subsequent layers, then ran Standard CJ on top of the ablated network. The pre-specified falsification criterion: if contact signal is localized to a single layer or layer band, we should see a non-monotonic profile (large drop at the localized depth, small drops elsewhere). Empirically the profile is **strictly monotonic** across the 16-layer window we tested (Table A9). Each ablation closer to mid-network removes incrementally more of the contact signal.

The interpretation: contact-relevant pair structure is built up progressively through the residual stream rather than localized at a single layer, with the late-layer band $\ell \in \{28, 32\}$ contributing the final ~ 5 pp of CJ’s contact precision. In-manifold mean replacement is cleaner than zero-ablation, which collapses the network into an out-of-distribution regime and gives uninformative drop magnitudes; we use mean replacement throughout.

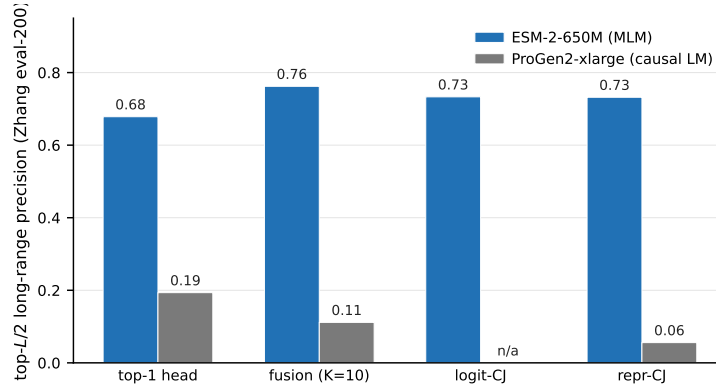


Figure A4: Contact-extraction methods on a causal LM (ProGen2-xlarge, next-token training) vs. a comparable masked language model (ESM-2-650M). Every method — top-1 attention head, naive-mean fusion at $K=10$, logit-CJ [Zhang et al., 2024], and repr-CJ (Section 3.2) — gives a substantially smaller top- $L/2$ long-range precision on ProGen2; logit-CJ is undefined for the causal LM and is shown as “n/a”.

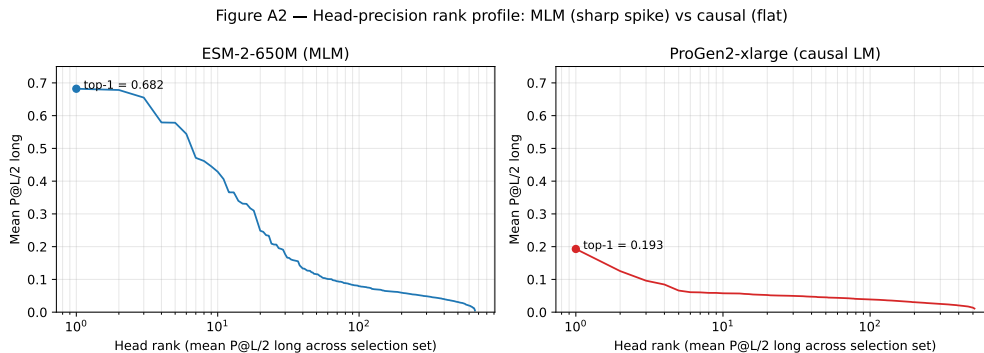


Figure A5: **Head-precision rank profile: MLM (sharp spike) vs causal (flat).** Mean $P@L/2$ long sorted by head rank, on log-rank axis (ESM-2-650M and ProGen2-xlarge, both on Zhang eval-200). ESM-2-650M shows the contact-head cluster spike; ProGen2-xlarge does not.

M Per-test wall-clock and selection-set transfer

Figure A6 shows the per-protein wall-clock vs. protein length L (panel a) and the amortized cost vs. test-set size N (panel b). Per-protein CJ wall scales with L while fusion stays roughly L -independent (its single forward dominates only at very large L). Table A10 reports the per-test wall-clock seconds, identification cost (10 selection-set forwards, matching the 10-protein selection budget used throughout), and amortized totals at $N=1000$ test proteins; the per-protein wall-clock numbers are medians measured across a representative Zhang timing sample (length range matched to eval-200) or 14–16 CASP14-FM evaluation runs, and are L -driven rather than dataset-specific.

Selection-set robustness. The per-architecture K choice is not overfit to the labeled selection set: Appendix N shows the top-10 head ranking is recovered from as few as 10 proteins with no downstream precision penalty (the per-architecture holdout- K transfer to a disjoint draw, omitted here for space, gave $\Delta_{\text{holdout-in-dist}} \geq 0$ on every cell that runs; the producing script is retained in the code release).

N Selection-set size sweep

How small can the labeled selection set get before the head ranking degrades? We subsample $N_{\text{sub}} \in \{3, 5, 10, 15, 20, 30, 50\}$ proteins from a 50-protein labeled pool (20 random draws per N_{sub} ;

Table A9: Depth-resolved mean-ablation probe on ESM-2-650M Zhang eval-200 ($N=200$). Single-layer = ablate residual at ℓ only; sustained = ablate at ℓ and re-apply at all subsequent layers. The two agree within 0.4pp on each row, so the monotonic accumulation reflects the network’s reliance on residual content at the ablation depth, not the propagation arithmetic of sustained replacement.

Ablation depth ℓ	Sustained CJ (P@L/2 long)	Single-layer CJ
unablated	0.7330	0.7330
$\ell = 32$	0.6801 (−5.3pp)	0.6811
$\ell = 28$	0.6704 (−6.3pp)	—
$\ell = 24$	0.6591 (−7.4pp)	—
$\ell = 16$	0.6046 (−12.8pp)	0.6032

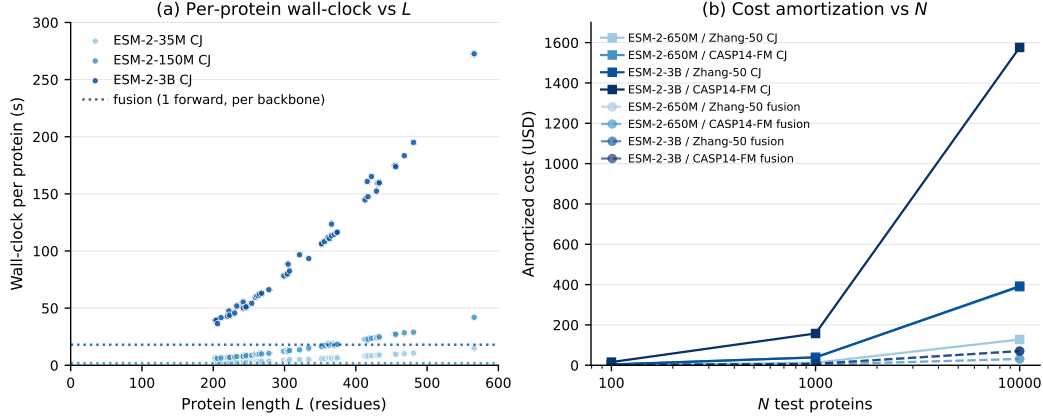


Figure A6: **Compute economy.** (a) Per-protein wall-clock vs. protein length L on ESM-2-35M, 150M, 3B: CJ markers scatter along the $O(L)$ trend, fusion (dotted horizontal) stays approximately constant at the per-forward cost of each backbone. (b) Amortized total cost vs. test-set size N on ESM-2- $\{650M, 3B\} \times \{Zhang, CASP14-FM\}$ (CJ solid, fusion dashed): CJ scales linearly with N ; fusion’s curve is dominated by the one-time 10-protein identification step at small N and barely moves thereafter. Break-even sits below $N=20$ on every cell measured. The curves are linear *extrapolations* from the measured per-protein costs of Table A10 (we did not run N to the thousands), so the large- N dollar figures — e.g. the ESM-2-3B/CASP14-FM CJ line passing \$1,600 by $N \approx 10k$ — are projected, not incurred.

deterministic at $N_{\text{sub}}=50$), rank heads by mean per-protein precision on the subsample, pick the top $K=10$, and evaluate naive-mean fusion on a disjoint 50-protein holdout. Table A11 reports the Jaccard overlap between the subsampled top-10 and the full-50 top-10, and the resulting holdout fusion precision.

The ranking is remarkably stable: at $N_{\text{sub}}=15$ every draw recovers the identical top-10 as the full 50-protein ranking (Jaccard = 1.0), and even at $N_{\text{sub}}=3$ the overlap averages 0.91 with no downstream precision penalty — the subsampled top-10 occasionally picks marginally different heads that score comparably on the holdout. The contact-head cluster is sufficiently prominent that a handful of labeled proteins identifies it. This sweep is what licenses the 10-protein selection set used for the in-distribution Zhang eval-200 results; the 50-protein budget elsewhere (the matched-budget L_1 -LR control) is conservative by comparison.

Table A10: Per-test wall and amortized cost at $N = 1000$. CJ s/prot is the measured mean wall-clock across the timing proteins (Zhang: representative length-matched sample; CASP14-FM: 14–16 runs). Fusion s/test is the measured median single-forward wall-clock for the test protein, fixed across datasets at each model scale because fusion is one forward per protein. Identification seconds is $10\times$ the per-protein forward time (the 10-protein selection budget). Dollar totals at $N = 1000$ are extrapolated linearly from these measured per-protein costs (CJ scales linearly in N , fusion scales as identification + $N\times$ per-test); the extrapolation assumes no further batching gains at large N .

Variant	Dataset	CJ s/prot	Fus s/test	Ident s	CJ \$ (N=1k)	Fus \$ (N=1k)
ESM-2-650M	Zhang	32.8	8.0	80	12.77	3.15
ESM-2-650M	CASP14-FM	100.0	8.0	80	38.89	3.15
ESM-2-3B	Zhang	100.9	18.0	180	39.25	7.07
ESM-2-3B	CASP14-FM	405.4	18.0	180	157.62	7.07

Table A11: **Selection-set size sweep on ESM-2-650M.** Jaccard overlap between the top- $K=10$ heads picked from a subsample of N_{sub} labeled proteins and from the full 50-protein set, plus the downstream fusion precision on a disjoint 50-protein holdout ($N=50$). Mean $\pm$ std over 20 random draws. **As few as 10–15 proteins recover the full-50 head ranking and downstream precision.**

N_{sub}	Jaccard vs full-50	Fusion P@ $L/2$ long	Range
3	0.909 ± 0.091	0.786 ± 0.005	[0.782, 0.795]
5	0.955 ± 0.079	0.784 ± 0.003	[0.782, 0.791]
10	0.991 ± 0.040	0.783 ± 0.002	[0.782, 0.791]
15	1.000 ± 0.000	0.782 ± 0.000	[0.782, 0.782]
20	0.991 ± 0.040	0.783 ± 0.002	[0.782, 0.791]
30	1.000 ± 0.000	0.782 ± 0.000	[0.782, 0.782]
50	1.000	0.782	—